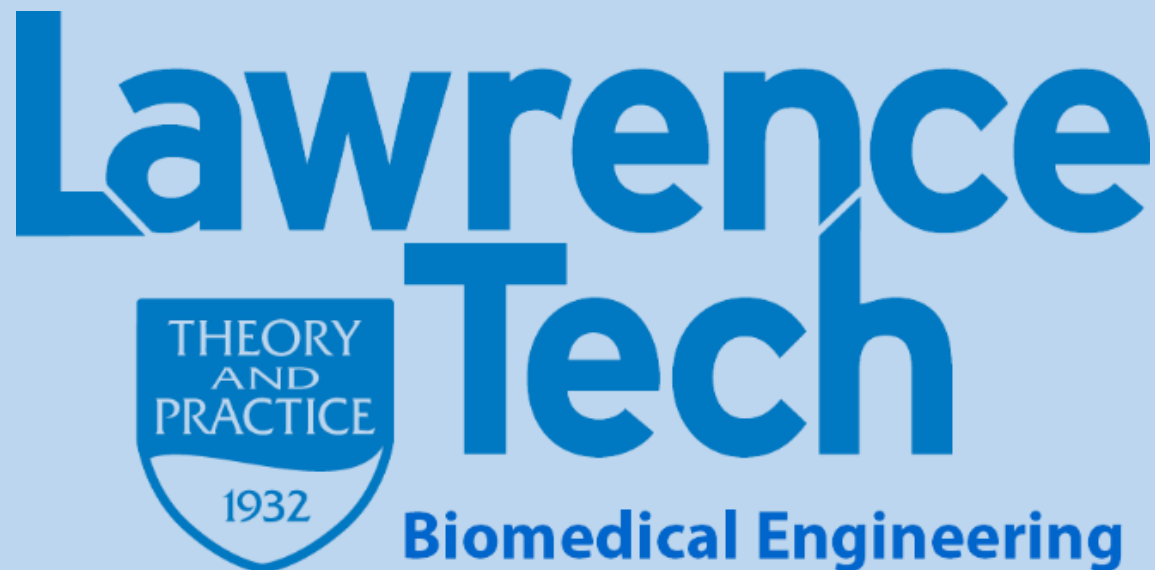




ShoulderX: Dynamic Band-Assisted Brace for Shoulder Joint Stability

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Abstract

To improve upon current shoulder brace designs focused on stabilizing the glenohumeral joint within the shoulder. Using TheraBands to produce stabilization with dynamic tension to increase shoulder strength and limit excess movement beyond maximum normal ranges of motion. This brace will serve as the mediator between injury and return to normal daily life.

Background

- The shoulder is the most mobile joint in the body, which naturally makes it prone to injuries like dislocation or subluxations.
- Overhead stability is vital when transitioning from injury back to daily activity. Achieving pain-free mobility and functional strength are key factors in long-term glenohumeral health.
- During the recovery phase comes a transition phase, where the patient's joint is encapsulated in a hard cast anymore, but still incapable of performing daily activities.



Figure 1: Shoulder Anatomy

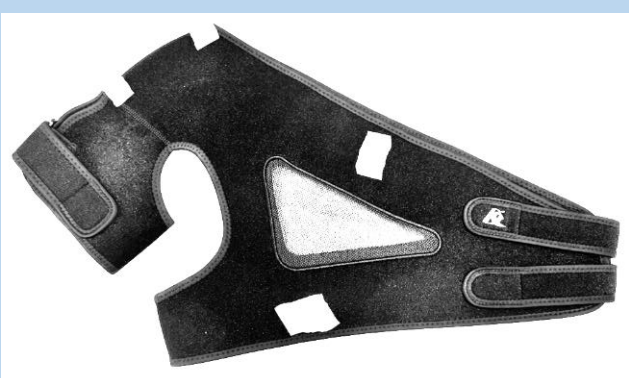


Figure 2: Overhead Flexion



Figure 3:

Design Process

Design Step/Prototype	Description of Step/Prototype
	TheraBands - A key aspect of ShoulderX. These bands are interchangeable and are what keeps the shoulder stable during flexion.
	Anaconda Brace - Our "Baseline" brace that we used to make a rough prototype from scratch.
	Addition of Therabands to Rough Prototype - Our team started finding ways to attach the bands to the rough prototype brace to see where they would have the most stabilization effect on the shoulder. Clips and buckles that were sewn on to attach them where needed.
	Addition of "Loops" and Change of "Base" Model - Loops were sewn where the white indicators are to ensure the bands don't move out of place when clipped on. This ensures the bands overlay the glenohumeral joint at all times during activity.
	Final Prototype - Our final design. The bands overlay the glenohumeral joint both vertically and horizontally to stabilize it. Additionally, the bands can be interchanged too for lighter or stronger resistibility.

Conclusion and Future Work

Conclusion

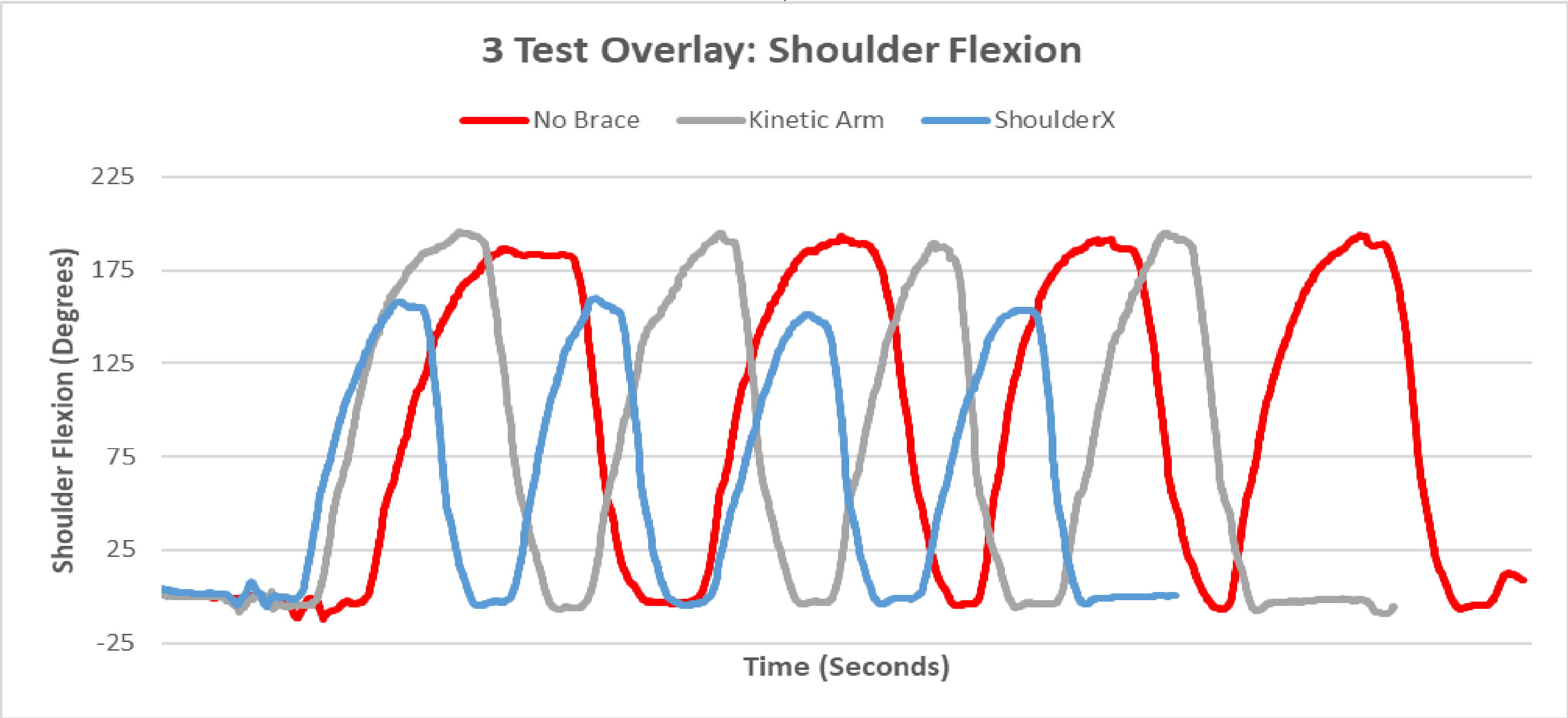
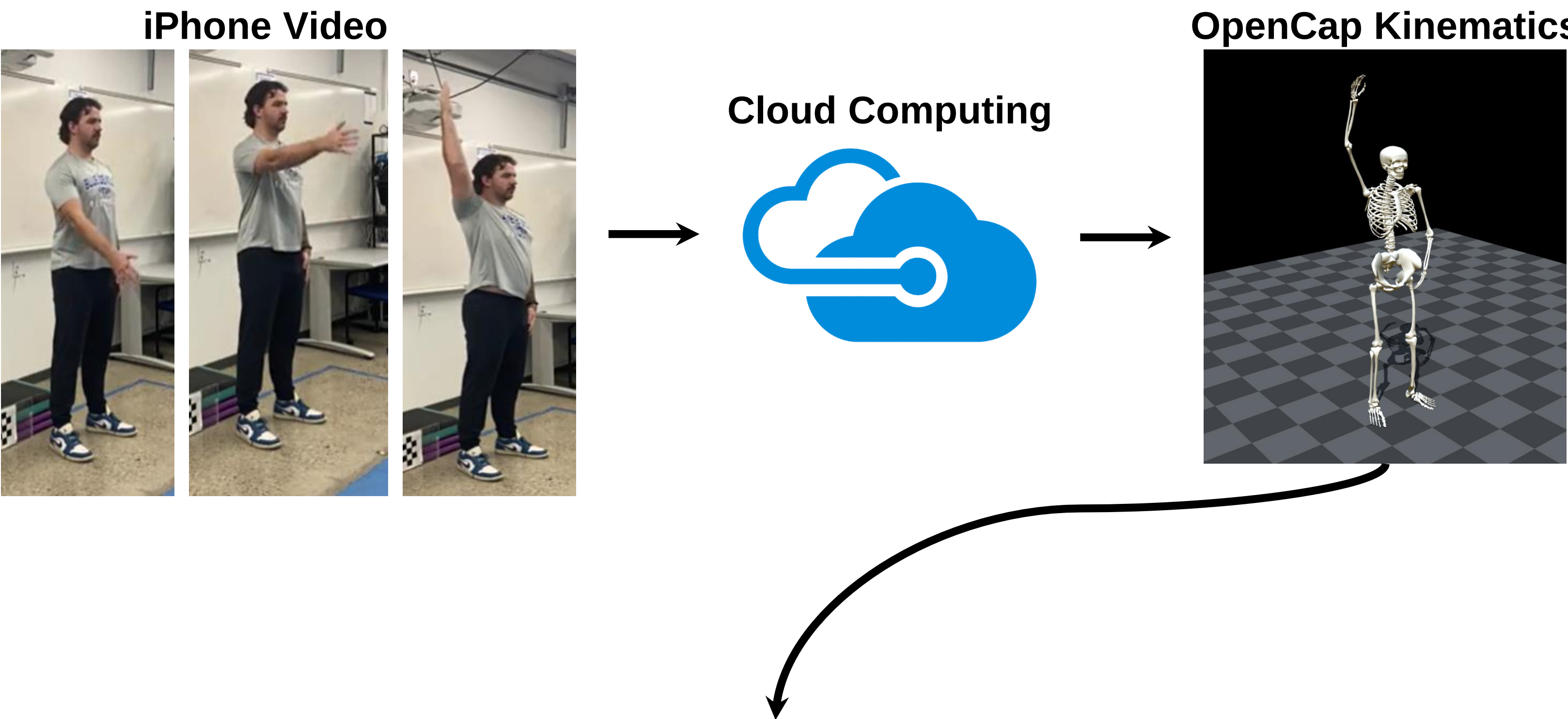
Our results suggest that ShoulderX improved shoulder stability during overhead flexion while still allowing functional movement. In addition to providing support to the glenohumeral joint, the brace was also perceived by users as secure, comfortable, and less bulky than other options tested. These findings indicate that ShoulderX may be effective not only in improving shoulder control during movement, but also in addressing the wearability concerns that often limit brace use. Overall, ShoulderX shows strong potential as a practical and user-friendly brace that balances stability, comfort, and mobility for active individuals recovering from shoulder instability.

Future Work

We plan to refine our dataset by testing ShoulderX on female subjects who are not overhead athletes, providing a necessary comparison to previous control groups. We also seek to expand our testing parameters by introducing varied TheraBand resistance levels and exploring alternative biomaterials to optimize the efficacy of the current elastic components.

Testing and Results

To evaluate the efficacy of the ShoulderX prototype in stabilizing the glenohumeral joint, comprehensive biomechanical testing was conducted within the Experimental Biomechanics Laboratory. We utilized OpenCap, an AI-driven, markerless motion capture system, to extract high-fidelity 3D kinematics during dynamic movement. The study evaluated a diverse cohort of 6 participants, comprising 4 overhead athletes (2 male, 2 female) and 2 non-overhead athletes (male). Each subject performed 4 continuous repetitions of overhead flexion. To establish a comparative performance profile, kinematics were recorded across three conditions: a control which is no brace, an industry standard (KineticArm), and the ShoulderX prototype. Furthermore, the ShoulderX was evaluated using distinct TheraBand stiffness levels to quantify how varying degrees of dynamic tension influence average peak shoulder flexion and overall joint stability.



OpenCap Data

*Kinematic data was processed via OpenCap's cloud-based musculoskeletal modeling, providing high-fidelity tracking of glenohumeral joint angles without the need for physical markers.

	NO BRACE		
Average Peak Shoulder Flexion (Degrees)	180.7°	176.3°	166.5°
Average Flexion Differential (Degrees)	N/A	4.38°	14.21°

User Feedback

User Feedback Results	Little to no pain for all test subjects during testing.	"I would say just the comfortability just because the other one felt like you were wearing a sleeve. Also being that your brace felt more secure for a shoulder injury as the kinetic arm seems like it wouldn't work as well."	"I liked the minimal look as it wasn't too bulky... it felt like it was actually supposed to help my shoulder stay stable." "It felt restrictive where it was supposed to, without being too tight or uncomfortable."
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The results suggest that ShoulderX improved shoulder stabilization during overhead flexion compared to the other test conditions while still maintaining functional mobility. This indicates that the brace was able to provide meaningful support to the glenohumeral joint without excessively limiting natural movement, which is especially important for individuals progressing from injury recovery back to normal activity or sport participation. In addition to its stabilizing effect, user feedback indicated that the brace was comfortable, secure, and less bulky than other options, highlighting its potential to improve both wearability and user compliance during repeated use. However, during testing there were some sizing errors (User to brace and vice versa), and there may also have been movement compensations during overhead flexion that influenced the results. These factors should be considered as limitations of the current testing process and suggest that further refinement and more controlled future testing will be important. Overall, the findings support the potential of ShoulderX as a functional and user-centered brace design that balances joint support, comfort, and performance needs, making it a promising option for active individuals and overhead athletes.

Acknowledgements

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